Established results and remaining scope

For each fixed strict Conway tree with sufficiently long, independently signed twists

Exact invariant formula

The full quantum sum, including every color, derivative term and normalization factor

Control of the full sum

Uniform bounds and a nonzero contribution near the geometric solution

Proved in this class

Actual knot volume

$$\lim_{N\to\infty} \frac{2\pi}{N} \log |K_N| = \mathrm{Vol}(S^3 \setminus K)$$

The main knot cusp stays unfilled

The same invariant from a pointed quantum A-operator

Data at finitely many fixed colors determine a period and a positive determinant (Gram) ratio.
Both have the proved knot-volume growth; the original peripheral factors are retained.

Remaining scope

Broader presentations and arbitrary short twists;
identifying the selected geometric factor of the classical A-polynomial.

These questions do not leave whole-sum control open for the covered class.

Earlier certified plat families are also established. No theorem for all knots is asserted.